PEER REVIEW HISTORY

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ARTICLE DETAILS

TITLE (PROVISIONAL)	Problems with Evidence Assessment in COVID-19 Health Policy Impact Evaluation: A systematic review of study design and evidence strength
AUTHORS	Haber, Noah; Clarke-Deelder, Emma; Feller, Avi; Smith, Emily; Salomon, Joshua; MacCormack-Gelles, Benjamin; Stone, Elizabeth M.; Bolster-Foucault, Clara; Daw, Jamie R.; Hatfield, Laura; Fry, Carrie E.; Boyer, Christopher B.; Ben-Michael, Eli; Joyce, Caroline M.; Linas, Beth S.; Schmid, Ian; Au, Eric; Wieten, Sarah; Jarrett, Brooke; Axfors, Cathrine; Nguyen, Van; Griffin, Beth; Bilinski, Alyssa; Stuart, Elizabeth A

VERSION 1 – REVIEW

REVIEWER	Alessandro Rovetta Mensana srls, Research and Disclosure Division
REVIEW RETURNED	19-Jul-2021

GENERAL COMMENTS	General comments Dear authors, thank you for the opportunity to read your interesting work. This review article systematically investigates the methodological quality of PubMed indexed publications regarding COVID-19 policy impact. To achieve this goal, the authors exploited a review tool based on a previous guide for COVID-19 policy impact based in turn on the PRISMA guidelines. The results support that much of the PubMed indexed literature is substantially unreliable on this issue. Therefore, the authors conclude that a combined peer-review process might perform better than the current one (based on separate single assessments). However, before proceeding with the publication, I believe that some points need to be properly addressed. ========== Major comments 1) Section: Methods - Review phases and procedures - Information Sources. “The study must be peer reviewed, and published in a peer-reviewed journal indexed by PubMed”. I suggest specifying whether the analysis includes MEDLINE and PMC or just MEDLINE. Furthermore, I suggest briefly discussing this choice. For example, why were other "prestigious" indices, such as Scopus (e.g., https://pubmed.ncbi.nlm.nih.gov/27053798/), not included? In particular, do the authors have evidence that the publication rate on PubMed was significantly higher than that on Scopus (or similar)? Or, is the intersection between PubMed indexing and Scopus (or others) large enough to well represent the literature indexed by both? Thank you.

	2) Section: Methods - Statistical analysis. 2.1. “Inter-rater reliability was assessed using Krippendorff’s alpha” The use of Krippendorff's alpha requires the verification of certain assumptions (https://www.tandfonline.com/doi/abs/10.1080/23808985.2013.11679142). I strongly suggest specifying if and how these have been verified. 2.2. “Journal impact factors were obtained from the 2019 Journal Citation Reports.[26]” I suggest motivating the use of the Impact Factor. In particular, a large literature disagrees with the adoption of this measure as it is subject to strong skewness and is sensitive to outliers (e.g., https://pubmed.ncbi.nlm.nih.gov/31745562/, https://www.nature.com/articles/466179b). 3) Section: Results - Descriptive Statistics. Alongside the median values, I strongly suggest reporting the interquartile ranges. 4) Section: Results - Review process assessment/Discussion. “This strongly suggests that the multiple reviewer review, discussion, and consensus process better identifies issues than independent review alone.” This finding is potentially very relevant but it should be appropriately deepened in the Discussion section to be fully justified. In particular, what are the advantages and limitations of a collaborative review? For example, is it not possible that, in this way, the most proficient oratorical reviewer will be able to influence colleagues? 5) Section: Discussion. “It is plausible and likely that other sets of reviewers would come to different conclusions.” This sentence is a sign of transparency and intellectual honesty. However, it raises a question that should be discussed in this manuscript: how can the lack of unambiguous judgment be addressed by the scientific community? Especially on the public health subject, such an issue is extremely relevant and deserves at least to be mentioned as a future research objective. In particular, theoretical discrepancies create security problems not only at the epidemiological but also infodemiological level, as they can significantly feed the distrust of the population in institutions and authorities (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6214269/, https://pubmed.ncbi.nlm.nih.gov/33370752/). 6) Section: Main Text/References. I suggest clearly specifying the cited papers that are preprint. This request is particularly relevant as the paper [18] is central to this study. ========== Minor comments 7) Section: References. Reference [17] has been published (https://pubmed.ncbi.nlm.nih.gov/34180960/). 8) Section: Table 1. I suggest distinguishing more the colors "No **" and "Mostly No". 9) Section: Discussion. “Not only are these assessments not the primary purpose of those studies, they also typically lack the detail

	requisite to make a critical assessment of the study design and methods used." This sentence is unwarranted unless motivated by a source.

REVIEWER	J. Rod Queensland University of Technology
REVIEW RETURNED	22-Jul-2021

GENERAL COMMENTS	Thanks for giving me the opportunity to review the manuscript. I really enjoyed reviewing the manuscript and have two main questions I would like the authors to discuss further given the importance of the topic. Overall comments: 1- I understand intuitively what you mean by "useful inference", but I'm having difficulties translating it to a cognitive scale of magnitudes. Perhaps a formal definition or introduction of what you understand by "useful inference" might help or existing examples of extreme cases that the authors know off that represent the poles of "useful inference". 2- I feel the authors are raising a valid issue, using appropriate methodology, and yielding important conclusions, but the focus of the paper seems to shift from policymaking to the academic community (please see the last comment about a "guide" to policymakers). More detailed comments: Introduction Line 33-34: Based on the age-adjusted infection fatality rate of covid-19, my impression is that the most important decisions were not related mainly to morbidity(in itself) and mortality, but to the potential of the virus to surpass worldwide health systems capacity and the economic trade-off to prevent a collapse of these systems. Line 18-19 page 6. Dot after reference. There are multiple instances with this happening. Every time the "outcome" of this review was presented I kept asking myself what is the author's scale for "useful inference". Whatever value of "usefulness" that the analysis yield, it is not mentioned whether it is of the same value for researchers and policymakers?. Although the paper is more focused on policymaking, the yielded "usefulness" results seem to be communicated for a research-oriented public. By "Minimal plausibility of useful and actionable evidence" I am guessing the authors mean the same as "useful inference"? if it is so, it would be probably better to use just one term. Discussion Line 27: Is there any included article within the review that did not have the mentioned failures and that you could provide as an example. This could help future policy analysis to overcome the potential highlighted issues?

	Line 29-30: I did not find claims of "citation-based metrics" implying "high-quality" in Yin et al 2021 paper. The only mention of "high-quality" is done in reference to peer-review and Yin et al 2021 acknowledged that this is not always the case. Yin et al 2021 did mention "high-impact" in regards to the number of citations but I did not find the authors suggesting high-impact = high-quality. I would suggest the authors think about rewriting the sentence suggesting Yin et al claims: citation metrics = high-quality, the statement seems unjustified. (If I miss it, I think it would perhaps be better to quote what Yin et al 2021 said). Line 31-32: I am having trouble with understanding the scale of "useful inference" implied in the sentence. Does the inclusion of your "Key design criteria" always imply useful inference? Line 22-23 page 17: "In other cases, it is plausible that many of these studies would not have been published had a more thorough or BETTER targeted methodological review been performed". I would suggest considering changing the word "better" for " a more" targeted... The use of the word better could imply that the previous peer-review process was "worse". I have the impression that the circumstances (knowledge-base, urgency, uncertainty, to name a few) surrounding the peer-review of policy interventions document during 2020 and the present document systematic peer-evaluation processes are not comparable. Thus I feel unease by the use of better/worse. I do agree though that the process was likely not as thorough as it could have been, but probably it was the best it could have been under the historical circumstances. In other words, I feel the point is well made even when removing "or BETTER targeted". The authors have validly painted a picture where the methodological quality of the evidence supporting policy action does not seem to follow "Key design criteria". This finding is used (also validly) to highlight uncertainty regarding the scientific support for policy action. They conclude "that we (I assumed here you mean policymakers in the context of the paper) often must make decisions when strong evidence is not feasible". However, I feel there is a comment missing in the discussion. Namely, a "guide" to policymakers about what to do in cases where strong evidence is not feasible. For example, I imagine the presented tool could perhaps be used to rank the methodological quality of the studies and help policymakers to give higher relevance to a particular set of studies that are more likely to be useful?. The figure presenting the information flow is not following the traditional PRISMA headlines: Identification, screening, eligibility, and inclusion.

REVIEWER	Zhaoli Dai Macquarie University, Australian Institute of Health Innovation
REVIEW RETURNED	26-Jul-2021

GENERAL COMMENTS	This systematic review conducted by Haber and colleagues aims to assess the strength of evidence in the published COVID-19 policy impact evaluation literature, using a review tool that was previously developed. They found 23/36 of studies analyzed the impact of stay-at-home requirements, and most studies did not “meet key design

	criteria" based on the assessment tool. After reviewing the details of the manuscript, I find that the study design, results presentation, and conclusion are generally vague. It is hard to understand what the study tells us about the lack of 'reliable' and "actionable evidence" in the COVID-19 policy implementation using this set of criteria, when we cannot relate these criteria to a typical policy evaluation framework. My specific comments are below. 1. Page 8, eligibility criteria: the covid-19 outcomes are not explicitly defined, line 19-20. What is included in the etc.? The authors should specify impact indicators for short-term and/or intermediate outcomes, as COVID-19 is an evolving viral infectious disease. Furthermore, as a pandemic, I am not sure whether economic impacts were qualified in this review. If not, perhaps the authors can provide a reason. 2. According to the response tracker by Oxford (https://www.bsg.ox.ac.uk/sites/default/files/2021-06/BSG-WP-2020-032-v12_0.pdf), the policy categories are more elaborate than what was included in this review. What struck me the most is the lack of studies on 'facial covering" and "protection of elderly people". Yet, there were 16/36 of the studies considered "other." The authors should include an explanation for this group. 3. Impact evaluation of policies relies on quantitative data. The authors used the review tool described to select studies on this aspect to draw a causal inference. In this aspect, I am not sure why cross-sectional studies would be qualified in the first place. Furthermore, what were the "others" mean that were not longitudinal observational studies or RCTs? However, on page 19 line 42-51, the authors mentioned that pre-post and RCTs were not qualified for the review. This seems to be inconsistent with the literature on the appropriate study designs for public policy assessments. 4. Page 11, it is confusing to me what the categories mean in terms of assessment of the evidence. It is unclear how mostly yes or mostly no were rated or unclear. For example, what proportion of the criteria met would be rated for these categories. 5. About the description of the process among the reviewers, please use a flow diagram. This would be easier for the readers to understand how the evidence was rated and a consensus was reached. 6. Page 17, in the Discussion, the authors summarized two primary problems of the evidence assessed- "problems are nearly ubiquitous throughout this literature: failure to isolate the impact of the policy(s) of interest from other changes that were occurring contemporaneously, and failure to appropriately address the functional form of infectious disease outcomes in a population setting." If there are multiple policy changes evaluated in a study, is there another set of method criteria to measure such impacts? For the second point, I am not sure what the authors were referring for "functional form of infectious disease." 7. Page 18, the authors acknowledge the tool " was limited to a very narrow - albeit critical - set of items." For the readers to understand this, the authors should consider explaining in more detail what each of these criteria means and how it is linked to the quantitative measure for drawing "causal inference." Also, it will be helpful to explain how this set of criteria is similar or different from other policy evaluation frameworks.

REVIEWER	Peng JIANG

	Sichuan University, Department of Industrial Engineering and Engineering Management
REVIEW RETURNED	12-Aug-2021

GENERAL COMMENTS	Comments for bmjopen-2021-053820. This is an interesting manuscript with a novel perspective for a systematic review of study design and evidence strength under the COVID-19 background. After going through the manuscript, I have some suggestions/comments to improve its quality. 1. As for the limitation, the authors thought that their review tool was limited to a very narrow set of items and the data quality. The reviewed papers were searched by November 2020. The readers might be curious about the overall time-serial variations in reviewed papers related to COVID-19 policies. Actually, the authors have the chance to add some latest papers and analyse the trends along with the COVID-19 development. 2. The “Methods” part is helpful for readers as this manuscript presents some new processes to evaluate the papers using expert experience. However, the “Methods” part can be placed into the appendix considering the length of this manuscript. This can also help readers focus more on the exciting findings/observations. 3. In the “Study Selection” part, eight reviewers screened the title and abstract of each article. Why are eight reviewers arranged for such a work? In addition, why not an odd number of reviewers? Generally, an odd number is more practical as the reviewers may offer conflicting opinions. 4. In the section of “Review tool for data collection”, “Finally, reviewers were asked a summary question: Overall: “Do you believe that the design is appropriate for identifying the policy impact(s) of interest?” This is quite a subjective question. However, the reviewer did not find any mathematical modelling regarding group decisions in this manuscript. 5. “The citation counts for accepted articles were obtained through Google Scholar on January 11”. However, the reviewed papers were not published same time. Therefore, the calculated statistics are not supportive. One option can be the citation counts of those papers within the same duration, e.g. half a year or one whole year after publication. 6. In the “Abstract” part, “outcomes published on November 26 or earlier and screening”. Please add the ‘year’ information, as it may confuse several years after the outbreak of the CIVOD-19 pandemic. 7. In the “Strengths and Limitations” part, “they may make other contributions to the literature.” The authors should make the contributions apparent.

VERSION 1 – AUTHOR RESPONSE

Reviewer: 1

Dr. Alessandro Rovetta, Mensana srls

Comments to the Author:

General comments

Dear authors, thank you for the opportunity to read your interesting work. This review article systematically investigates the methodological quality of PubMed indexed publications regarding

COVID-19 policy impact. To achieve this goal, the authors exploited a review tool based on a previous guide for COVID-19 policy impact based in turn on the PRISMA guidelines. The results support that much of the PubMed indexed literature is substantially unreliable on this issue. Therefore, the authors conclude that a combined peer-review process might perform better than the current one (based on separate single assessments). However, before proceeding with the publication, I believe that some points need to be properly addressed.

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Major comments

1) Section: Methods - Review phases and procedures - Information Sources. “The study must be peer reviewed, and published in a peer-reviewed journal indexed by PubMed”. I suggest specifying whether the analysis includes MEDLINE and PMC or just MEDLINE. Furthermore, I suggest briefly discussing this choice. For example, why were other "prestigious" indices, such as Scopus (e.g., https://pubmed.ncbi.nlm.nih.gov/27053798/), not included? In particular, do the authors have evidence that the publication rate on PubMed was significantly higher than that on Scopus (or similar)? Or, is the intersection between PubMed indexing and Scopus (or others) large enough to well represent the literature indexed by both? Thank you.

Thank you for the comment. We had written the main reasoning below (prominence and near-comprehensiveness). One key reason here is that we believe that the expected returns for expanding to other indexing services would be low, such that it would provide minimal additional articles at increased logistical and screening cost. We have expanded upon our original text explaining this decision as below:

> **PubMed was chosen as the exclusive indexing source due to the prevalence and prominence of policy impact studies in the health and medical field. Pubmed *is a fairly comprehensive list of the peer-reviewed journals likely to publish these works, and focusing only on this source helped to keep screening time and logistics more manageable.***

2) Section: Methods - Statistical analysis.

2.1. “Inter-rater reliability was assessed using Krippendorff’s alpha” The use of Krippendorff's alpha requires the verification of certain assumptions (https://www.tandfonline.com/doi/abs/10.1080/23808985.2013.11679142). I strongly suggest specifying if and how these have been verified.

We agree that Krippendorff's alpha (KA) typically relies on a number of assumptions, and we thank the reviewer for pointing this out. We use Krippendorff's alpha because we have many reviewers, an ordinal scale, and "missing" values. Moreoever, this metric is commonly used to compare results across coders with similar data. We prefer a metric that accommodates these features and are willing to accept the additional assumptions. However, as is true for many statistical measures, the assumptions underlying Krippendorff's alpha are not easily verifiable (e.g. the trinity distribution assumption), as indicated in the (now cited) Zhao et al., 2013, nor are they particularly impactful on the interpretation of our results if violated.

Rather than more precise typical uses of inter-rater reliability (in which a study design is intended with the explicit goal of having high IRR such that KA is a test of validity), Krippendorff's alpha in this paper is being used informally to indicate heterogeneity at the independent phase. In this case, violations of the metric's assumptions would not substantially impact our finding of high heterogeneity.

To help make this clearer, we have also added a new metric which simply sums the categorical differences between reviewers for main outcomes between primary reviewers, to further demonstrate the heterogeneity, and have clarified how we are using KA and why.

> **Inter-rater reliability (IRR) was assessed using Krippendorff's alpha.** ***Rather than more typical uses intended as an examination of the "validity" of ratings, the IRR statistic in this case is being used as a heuristic indicator of heterogeneity between reviewers during the independent phase, where heterogeneity is both expected and not necessarily undesirable. As a second examination of reviewer heterogeneity, we also show the distribution of category differences between primary reviewers within a study (e.g. if primary reviewers rated "Yes," "Mostly no," and "Mostly yes" there are two pairs of answers that were one category different, and one pair that was two categories different).***
>
> ***…***
>
> ***For the main summary rating, primary reviewers within each study agreed in 26% of cases (n=16), were one category different in 45% (n=46), two categories different in 19% (n=12), and three categories (i.e. the maximum distance, "Yes" vs "No") in 10% of cases (n=6).***

2.2. "Journal impact factors were obtained from the 2019 Journal Citation Reports.[26]" I suggest motivating the use of the Impact Factor. In particular, a large literature disagrees with the adoption of this measure as it is subject to strong skewness and is sensitive to outliers (e.g., https://pubmed.ncbi.nlm.nih.gov/31745562/, https://www.nature.com/articles/466179b).

We agree that Journal Impact Factor is a flawed measure, and in this study we don't find that Journal Impact Factor is a reliable metric of quality; if anything, our research demonstrates the opposite. We use the Impact Factor and citation counts to describe the journals in which the included articles were published. This helps readers understand that the included articles are from high-impact, highly visible literature. Impact factor is never used as a control variable, and no associations with it are run.

3) Section: Results - Descriptive Statistics. Alongside the median values, I strongly suggest reporting the interquartile ranges.

Interquartile ranges have now been reported, as below:

> **Publication information from our sample is shown in Figure 2. The articles in our sample were generally published in journals with high impact factors (median impact factor: 3.6*, 25th percentile:* 2.3*, 75th percentile:* 5.3 *IQR: 3.0*) and have already been cited in the academic literature (median citation count: 5.0*, 25th percentile: 2.0, 75th percentile: 26.8, IQR 24.8,* on 1/11/21).**

4) Section: Results - Review process assessment/Discussion. "This strongly suggests that the multiple reviewer review, discussion, and consensus process better identifies issues than independent review alone." This finding is potentially very relevant but it should be appropriately deepened in the Discussion section to be fully justified. In particular, what are the advantages and limitations of a collaborative review? For example, is it not possible that, in this way, the most proficient oratorical reviewer will be able to influence colleagues?

As both authors of this manuscript and participants in the review process, we found that the interactive, collaborative, iterative review process identified issues better than our initial independent reviews, as indicated both by our collective experiences performing the review and by elicited feedback to the administrators. On numerous occasions during the collaborative phase, we found that our fellow reviewers discovered issues we had missed and contributed complementary expertise that both deepened and broadened our understanding of the methodological issues in the papers we reviewed.

To clarify this (note also the overlap with comment #5 below), we have added the following:

> ***While subjective assessments are inherently subject to the technical expertise, experiences, and opinions of reviewers, we argue they are both appropriate and necessary to reliably assess strength of evidence based on theoretical methodological issues. With the exception of the graphical assessment, proper assessment of the core methodological issues requires that reviewers are able to weigh the evidence as they see fit.*** **Much like**

standard institutional peer review, reviewers independently had highly heterogeneous opinions, attributable to differences in opinion *or training*, misunderstandings/learning about the review tool and process, and expected reliance on the consensus process. Unlike traditional peer review, there was *subject-matter-specific guidance and a* process to consolidate and discuss those heterogenous initial opinions. *The reduction in ratings from the initial highly heterogeneous ratings to a lower heterogeneity in ratings indicates that reviewers had initially identified issues differently, but that the discussion and consensus process helped elucidate the extent of the different issues that each reviewer detected and brought to discussion. This also reflects reviewer learning over time, where reviewers were better able to identify issues at the consensus phase than earlier. It is plausible that stronger opinions had more weight, but we expect that this was largely mitigated by the random assignment of the arbitrator, and reviewer experiences did not indicate this as an issue.*

5) Section: Discussion. “It is plausible and likely that other sets of reviewers would come to different conclusions.” This sentence is a sign of transparency and intellectual honesty. However, it raises a question that should be discussed in this manuscript: how can the lack of unambiguous judgment be addressed by the scientific community? Especially on the public health subject, such an issue is extremely relevant and deserves at least to be mentioned as a future research objective. In particular, theoretical discrepancies create security problems not only at the epidemiological but also infodemiological level, as they can significantly feed the distrust of the population in institutions and authorities (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6214269/, https://pubmed.ncbi.nlm.nih.gov/33370752/).

Thank you for these thoughtful comments on the implications of our work for the review process. To more thoroughly address this and other reviewer comments, we have moved a few items around and written a more comprehensive discussion of the consensus process as below:

This systematic review of evidence strength has limitations. The tool itself was limited to a very narrow - albeit critical - set of items. Low ratings in our study should not be interpreted as being overall poor studies, as they may make other contributions to the literature that we did not evaluate. While the *guidance and tool* provided a well-structured framework and our reviewer pool was well-qualified, strength of evidence review is inherently subjective. It is plausible and likely that other sets of reviewers *could* come to different conclusions *for each study, but unlikely that the overall conclusions of our assessment would change substantially.* However, the consensus process was designed with *subjectivity* in mind, and *our study* demonstrates the value of consensus processes for overcoming hurdles with subjective and difficult decisions.

***While subjective assessments are inherently subject to the technical expertise, experiences, and opinions of reviewers, we argue they are both appropriate and necessary to reliably assess strength of evidence based on theoretical methodological issues. With the exception of the graphical assessment, proper assessment of the core methodological issues requires that reviewers are able to weigh the evidence as they see fit.* Much like standard institutional peer review, reviewers independently had highly heterogeneous**

opinions, attributable to differences in opinion *or training*, misunderstandings/learning about the review tool and process, and expected reliance on the consensus process. Unlike traditional peer review, there was *subject-matter-specific guidance and a* process to consolidate and discuss those heterogenous initial opinions. *The reduction in ratings from the initial highly heterogeneous ratings to a lower heterogeneity in ratings indicates that reviewers had initially identified issues differently, but that the discussion and consensus process helped elucidate the extent of the different issues that each reviewer detected and brought to discussion. This also reflects reviewer learning over time, where reviewers were better able to identify issues at the consensus phase than earlier. It is plausible that stronger opinions had more weight, but we expect that this was largely mitigated by the random assignment of the arbitrator, and reviewer experiences did not indicate this as an issue.*

We hope that this allays concerns

6) Section: Main Text/References. I suggest clearly specifying the cited papers that are preprint. This request is particularly relevant as the paper [18] is central to this study.

Our review only included peer-reviewed articles. For all references, we will conform to BMJ standards on formatting and citation of references. Reference 18 is the protocol pre-registration for our study, rather than a preprint.

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Minor comments

7) Section: References. Reference [17] has been published (https://pubmed.ncbi.nlm.nih.gov/34180960/).

Thank you for catching this. Final acceptance of this article occurred after ours had been submitted, and we have updated the citation.

8) Section: Table 1. I suggest distinguishing more the colors "No **" and "Mostly No".

We have shifted the colors slightly to distinguish these more easily.

9) Section: Discussion. “Not only are these assessments not the primary purpose of those studies, they also typically lack the detail requisite to make a critical assessment of the study design and methods used.” This sentence is unwarranted unless motivated by a source.

We have removed this sentence

Reviewer: 2

Dr. J. Rod, Queensland University of Technology

Comments to the Author:

Thanks for giving me the opportunity to review the manuscript.

I really enjoyed reviewing the manuscript and have two main questions I would like the authors to discuss further given the importance of the topic.

Overall comments:

1- I understand intuitively what you mean by "useful inference", but I'm having difficulties translating it to a cognitive scale of magnitudes. Perhaps a formal definition or introduction of what you understand by "useful inference" might help or existing examples of extreme cases that the authors know off that represent the poles of "useful inference".

In this case, we are framing the argument as minimal criteria for possibly useful (or actionable) inference. We believe that evidence is useful if it would be valuable to a future decision-maker by reducing the uncertainty around some parameter that influences their decision. In the context of our study, inference that does not identify the causal effect of the policy is not actionable, and therefore not particularly “useful.” We have clarified this with the language below:

> ***For this study, we ascertain whether studies are able to plausibly identify causal effects of policies, based on criteria meant to identify information of greatest value to inform policy decisions. The causal estimand is something that, if known, would definitely help policy makers decide what to do (e.g., whether to implement or discontinue a policy). The study estimates that target causal impact with a rigorous design and appropriate data in a relevant population. For shorthand, we refer to this set of criteria as “actionable” evidence.***

2- I feel the authors are raising a valid issue, using appropriate methodology, and yielding important conclusions, but the focus of the paper seems to shift from policymaking to the academic community (please see the last comment about a "guide" to policymakers).

At this stage, we are speaking mostly to the academic community who produced, vetted, and distributed these policy analysis papers, many of which have been cited and used by policy decision-makers. Some of our main action recommendations are for the academic community to better vet important studies such that decision-makers have clearer signals of what is robust and "useful" information from which to draw. We address the policymakers portion of this question below.

More detailed comments:

Introduction

Line 33-34: Based on the age-adjusted infection fatality rate of covid-19, my impression is that the most important decisions were not related mainly to morbidity(in itself) and mortality, but to the potential of the virus to surpass worldwide health systems capacity and the economic trade-off to prevent a collapse of these systems.

There are many potential goals for policy related to mitigating the potential harms of COVID-19. Certainly potential issues such as "long COVID" might be a concern to policy makers. We are not attempting to assess what those tradeoffs might be or the motivations behind policies, but rather our ability to assess policy impacts from published studies.

Line 18-19 page 6. Dot after reference. There are multiple instances wtih this happening.

We believe that we have complied with the formatting, but in any case we will work with the BMJ Open copyediting team to ensure that all references and formatting meet expectations.

Every time the "outcome" of this review was presented I kept asking myself what is the author's scale for "useful inference". Whatever value of "usefulness" that the analysis yield, it is not mentioned whether it is of the same value for researchers and policymakers?. Although the paper is more focused on policymaking, the yielded "usefulness" results seem to be communicated for a research-oriented public.

By "Minimal plausibility of useful and actionable evidence" I am guessing the authors mean the same as "useful inference"? if it is so, it would be probably better to use just one term.

We agree that this was unclear, and thank you for your comments. We changed the language here to be much clearer, including giving clearer definitions of what we mean. We have made a number of changes, starting with changing the terminology to be "actionable" throughout for clarity rather than "useful." Note that we do not fully define what is actionable (no definition can fully encompass all dimensions of what makes a study actionable), but rather define what properties of a study make it *in*actionable.

We have made a number of small changes throughout (removing the word "useful" in particular"), and have added the following section:

> ***For this study, we ascertain whether studies are able to plausibly identify causal effects of policies, based on criteria meant to identify information of greatest value to inform policy decisions. The causal estimand is something that, if known, would definitely help policy makers decide what to do (e.g., whether to implement or discontinue a policy). The study estimates that target causal impact with a rigorous design and appropriate data in a relevant population. For shorthand, we refer to this set of criteria as "actionable" evidence.***

Discussion

Line 27: Is there any included article within the review that did not have the mentioned failures and that you could provide as an example. This could help future policy analysis to overcome the potential highlighted issues?

One study was rated as not having failures in any of our criteria. However, rather than highlight it in particular, we prefer to make a more general comment about the selection of scenarios.

> **The framework for the review tool is based on the requirements and assumptions built into policy evaluation methods. Quasi-experimental methods rely critically on the scenarios in which the data are generated. These assumptions and the circumstances in which they are plausible are well-documented and understood,[2,4–6,17,65] including one paper discussing application of difference-in-differences methods specifically for COVID-19 health policy, released in May 2020.[5] While "no uncontrolled concurrent changes" is a difficult bar to clear, that bar is fundamental to inference using these methods. *The feasibility of producing actionable evidence relies critically on the particular scenario in which the data were generated. The key to surpassing this issue relies on identifying the right scenarios for actionable inference, and being extremely transparent when we are not in those scenarios.***

Line 29-30: I did not find claims of "citation-based metrics" implying "high-quality" in Yin et al 2021 paper. The only mention of "high-quality" is done in reference to peer-review and Yin et al 2021 acknowledged that this is not always the case. Yin et al 2021 did mention "high-impact" in regards to the number of citations but I did not find the authors suggesting high-impact = high-quality. I would suggest the authors think about rewriting the sentence suggesting Yin et al claims: citation metrics = high-quality, the statement seems unjustified. (If I miss it, I think it would perhaps be better to quote what Yin et al 2021 said).

Thank you for your careful eye here. The particular phrase "high-quality" is as you suggest, although there are many references to examining "quality" throughout the paper. We have clarified the statement as below:

> **While policy decisions are being made on the backs of high impact-factor papers, we find that the citation-based metrics do not correspond to "quality" research as *used by* Yin et al., 2021.[63]**

Line 31-32: I am having trouble with understanding the scale of "useful inference" implied in the sentence. Does the inclusion of your "Key design criteria" always imply useful inference?

We discuss this a bit more above. Meeting these criteria does not imply that inference is useful (as that will depend on a number of factors including the research question itself), but failing to meet these criteria does indicate that this inference is not actionable.

Line 22-23 page 17: "In other cases, it is

plausible that many of these studies would not have been published had a more thorough or BETTER targeted methodological review been performed". I would suggest considering changing the word "better" for " a more" targeted... The use of the word better could imply that the previous peer-review process was "worse". I have the impression that the circumstances (knowledge-base, urgency, uncertainty, to name a few) surrounding the peer-review of policy interventions document during 2020 and the present document systematic peer-evaluation processes are not comparable.

Thus I feel unease by the use of better/worse. I do agree though that the process was likely not as thorough as it could have been, but probably it was the best it could have been under the historical circumstances. In other words, I feel the point is well made even when removing "or BETTER targeted".

This is an excellent point. We have revised to be clearer.

> **At minimum, the flaws and limitations in their inference could have been communicated at the time of publication, when they are needed most. In other cases, it is plausible that**

many of these studies would not have been published had a more thorough or *more* targeted methodological review been performed.

...

This strongly suggests that the multiple reviewer review, discussion, and consensus process *more thoroughly* identifies issues than independent review alone.

The authors have validly painted a picture where the methodological quality of the evidence supporting policy action does not seem to follow "Key design criteria". This finding is used (also validly) to highlight uncertainty regarding the scientific support for policy action. They conclude "that we (I assumed here you mean policymakers in the context of the paper) often must make decisions when strong evidence is not feasible". However, I feel there is a comment missing in the discussion. Namely, a "guide" to policymakers about what to do in cases where strong evidence is not feasible.

We agree that we would love to be able to thoroughly provide policy guidance for making decisions under uncertainty. However, this issue encompasses an entire field (decision science), and we are unfortunately unable to give this topic adequate detail in order to be sufficiently informative. We have added the following information to the discussion clearer:

For policymakers, it can be difficult to make decisions in a setting with both high uncertainty and a proliferation of research of varying quality. We recommend consulting with researchers with the training to discern low from high quality research and adopting techniques to consider how uncertainty shapes the range of possible outcomes from a decision.

For example, I imagine the presented tool could perhaps be used to rank the methodological quality of the studies and help policymakers to give higher relevance to a particular set of studies that are more likely to be useful?.

Unfortunately, we do not believe that rank ordering would be valuable in this case, particularly given that the strength of evidence was so low in this sample. As discussed in the discussion section, this study is merely a first pass for being plausibly actionable, and further and more detailed review is needed for the few studies that passed these minimal criteria. This tool is intended as a minimal set of criteria, rather than a scoring system, similar to recommendations for other tools such as the Risk of Bias tool (see below for additional changes).

The figure presenting the information flow is not following the traditional PRISMA headlines: Identification, screening, eligibility, and inclusion.

Thank you for pointing this out. We have revised our headings to be more in line with the PRISMA guidelines here: http://prisma-statement.org/prismastatement/flowdiagram.aspx

Reviewer: 3

Dr. Zhaoli Dai, Macquarie University

Comments to the Author:

This systematic review conducted by Haber and colleagues aims to assess the strength of evidence in the published COVID-19 policy impact evaluation literature, using a review tool that was previously developed. They found 23/36 of studies analyzed the impact of stay-at-home requirements, and most studies did not "meet key design criteria" based on the assessment tool.

After reviewing the details of the manuscript, I find that the study design, results presentation, and conclusion are generally vague. It is hard to understand what the study tells us about the lack of 'reliable' and "actionable evidence" in the COVID-19 policy implementation using this set of criteria, when we cannot relate these criteria to a typical policy evaluation framework. My specific comments are below.

1. Page 8, eligibility criteria: the covid-19 outcomes are not explicitly defined, line 19-20. What is included in the etc.? The authors should specify impact indicators for short-term and/or intermediate outcomes, as COVID-19 is an evolving viral infectious disease. Furthermore, as a pandemic, I am not sure whether economic impacts were qualified in this review. If not, perhaps the authors can provide a reason.

Thank you for pointing out the lack of clarity. We have clarified the inclusion criteria as below to more thoroughly address this. We do not specify the time-frames for the outcomes, as we would prefer to address all studies meeting our criteria, regardless of the timeframes. The "vagueness" is in part because it is necessary to balance reviewer judgements and assessments, as these are not all clear cut. We have clarified the inclusion criteria below.

- **The primary topic of the article must be evaluating one or more individual COVID-19 or *SARS-CoV-2* policies on direct COVID-19 or *SARS-CoV-2* outcomes**
    - **The primary exposure(s) must be a policy, defined as a government-issued order at any government level to address a directly COVID-19-related outcome (e.g., mask requirements, travel restrictions, etc).**
    - ***Direct* COVID-19 or *SARS-CoV-2* outcomes *are those that are specific to disease and health outcomes* may include cases detected, mortality, number of tests taken, test positivity rates, Rt, etc.**
    - **This may NOT include indirect impacts of COVID-19 on items that are not direct *COVID-19 or SARS-CoV-2 impacts* such as income, childcare, *economic impacts, beliefs and attitudes, etc.***
- **The primary outcome being examined must be a COVID-19-specific outcome, as above.**

- **The study must be designed as an impact evaluation study from primary data (i.e., not primarily a predictive or simulation model or meta-analysis).**
- **The study must be peer reviewed, and published in a peer-reviewed journal indexed by PubMed.**
- **The study must have the title and abstract available via PubMed at the time of the study start date (November 26).**
- **The study must be written in English.**

2. According to the response tracker by Oxford (https://www.bsg.ox.ac.uk/sites/default/files/2021-06/BSG-WP-2020-032-v12_0.pdf), the policy categories are more elaborate than what was included in this review. What struck me the most is the lack of studies on 'facial covering" and "protection of elderly people". Yet, there were 16/36 of the studies considered "other." The authors should include an explanation for this group.

The categorization used in this study was the most up-to-date categorization used by the Oxford Government Response Tracker at the time of the conduct of this study (November 2020-January 2021). They later revised their categorizations.

We have additionally added the following text to clarify what is in the "other" category:

> ***Reviewers specified mask mandates for 3 of the studies, and noted either a combination of many interventions or unspecified specific policies in 7 cases.***

3. Impact evaluation of policies relies on quantitative data. The authors used the review tool described to select studies on this aspect to draw a causal inference. In this aspect, I am not sure why cross-sectional studies would be qualified in the first place. Furthermore, what were the "others" mean that were not longitudinal observational studies or RCTs? However, on page 19 line 42-51, the authors mentioned that pre-post and RCTs were not qualified for the review. This seems to be inconsistent with the literature on the appropriate study designs for public policy assessments.

Thank you for the comments. As we describe in the methods section, cross-sectional, pre-post, and RCTs are all explicitly counted as meeting our inclusion criteria and are included in the sample, but do not undergo full review. Both cross-sectional designs and pre/post designs do not meet our criteria for actionable evidence per the impact evaluation guidance and undergo no further review. In our guidance document, we explain the rationale behind why pre/post studies cannot yield well-identified causal inference in the specific context of COVID-19 policy impact evaluation, including reference to related text in the area.

RCTs would meet our criteria, but do not receive further review because our review questions are focused on assessing whether the assumptions required for causal inference from observational data are met. However, no RCTs were identified at the time of this research, as shown in the Results section.

We have added the following:

> **Studies categorized as cross-sectional, pre/post, randomized controlled trial designs were *included in our sample, but* set aside for no further review for the purposes of this research. …. *Cross-sectional and pre/post studies are not considered sufficient to yield well-identified causal inference in the specific context of COVID-19 policy impact evaluation, as explained in the policy impact evaluation guidance documentation [18].***

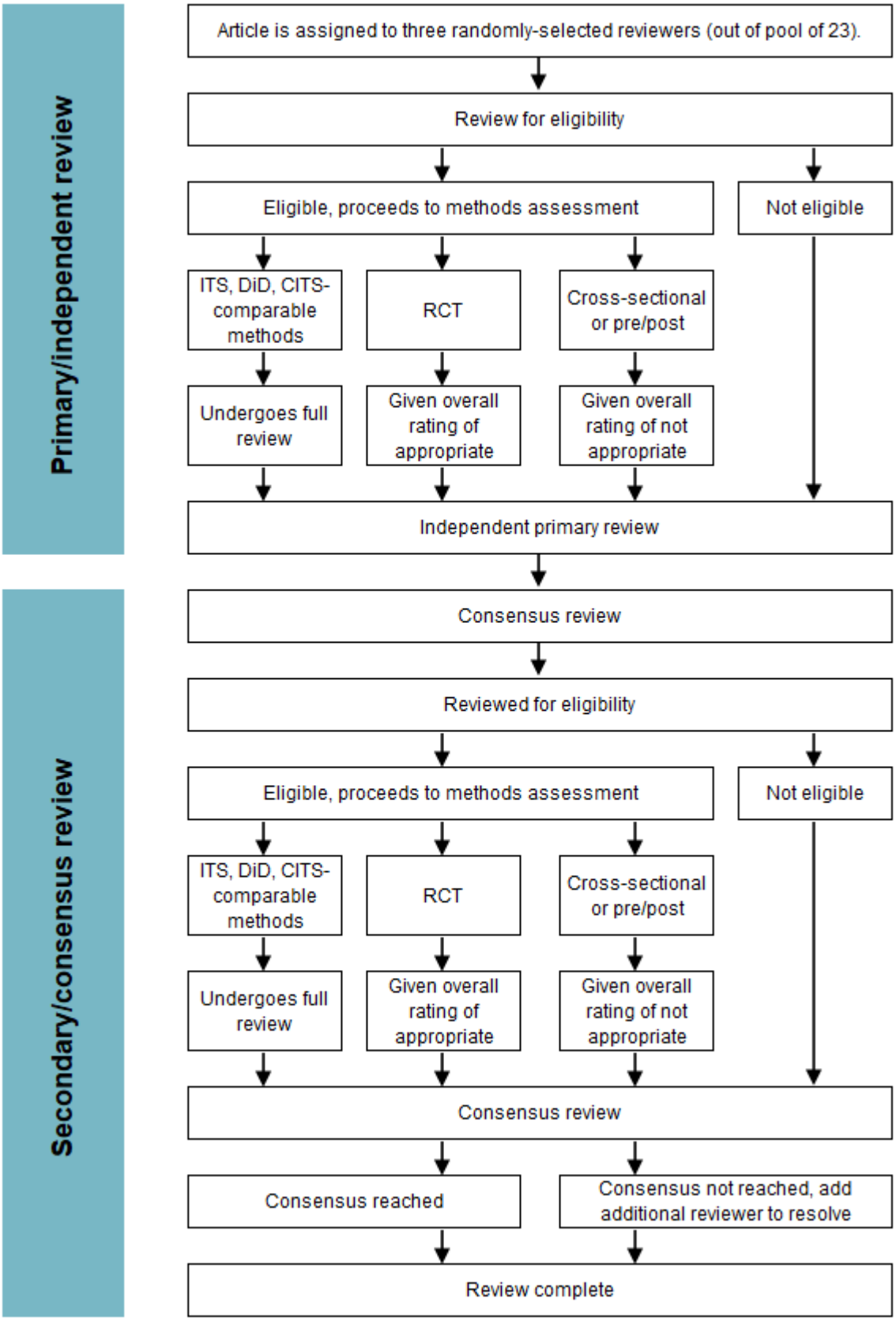
4. Page 11, it is confusing to me what the categories mean in terms of assessment of the evidence. It is unclear how mostly yes or mostly no were rated or unclear. For example, what proportion of the criteria met would be rated for these categories.

This was, by design, a subjective assessment of the totality of the question. Numerical indices are broadly inappropriate for strength of evidence assessments, and comprehensive subjective assessment is necessary to evaluate strength of evidence as a whole. This is similar to advice from other tools. For example, even the new algorithms for scoring in the new RoB 2 tool from Cochrane only suggest that these algorithms are only "proposed" judgements, requiring that analysts verify or make changes to these as appropriate (https://training.cochrane.org/handbook/current/chapter-08). It is entirely plausible, for example, for a study to meet all but one criteria but for the one remaining to be sufficiently violated that the entire collective category is compromised. Alternatively, there could be many minor violations of all of the criteria, but that they were collectively not sufficiently problematic to impact overall ratings.

> **For each of the four key questions, reviewers were given the option to select "No," "Mostly no," "Mostly yes," and "Yes" with justification text requested for all answers other than "Yes." Each question had additional prompts as guidance, and with much more detail provided in the full guidance document. *Ratings are, by design, subjective assessments of the category according to the guidance. We do not use a numerical scoring, for similar reasons as Cochrane suggests that the algorithms for summary judgements for the RoB2 tool are merely "proposed" assessments, which reviewers should change as they believe appropriate. It is entirely plausible, for example, for a study to meet all but one criteria but for the one remaining to be sufficiently violated that the entire collective category is compromised. Alternatively, there could be many minor violations of all of the criteria, but that they were collectively not sufficiently problematic to impact overall ratings. Further, reviewers were also tasked with considering room for doubt in cases where answers to these questions were unclear.***

5. About the description of the process among the reviewers, please use a flow diagram. This would be easier for the readers to understand how the evidence was rated and a consensus was reached.

We have now created a flow diagram to make the process clearer, as below, and referred to it in the paper. This is in Appendix 3.


Primary/independent review
Article is assigned to three randomly-selected reviewers (out of pool of 23).
Review for eligibility
Eligible, proceeds to methods assessment
Not eligible
ITS, DiD, CITS-comparable methods
RCT
Cross-sectional or pre/post
Undergoes full review
Given overall rating of appropriate
Given overall rating of not appropriate
Independent primary review
Secondary/consensus review
Consensus review
Reviewed for eligibility
Eligible, proceeds to methods assessment
Not eligible
ITS, DiD, CITS-comparable methods
RCT
Cross-sectional or pre/post
Undergoes full review
Given overall rating of appropriate
Given overall rating of not appropriate
Consensus review
Consensus reached
Consensus not reached, add additional reviewer to resolve
Review complete

6. Page 17, in the Discussion, the authors summarized two primary problems of the evidence assessed- “problems are nearly ubiquitous throughout this literature: failure to isolate the impact of the policy(s) of interest from other changes that were occurring contemporaneously, and failure to appropriately address the functional form of infectious disease outcomes in a population setting.” If there are multiple policy changes evaluated in a study, is there another set of method criteria to measure such impacts?

The same criteria were used to evaluate packages of policies and individual policies. The strength of evidence does not depend on the exact content of the policy. However, we evaluated whether the authors accounted for the potential impact of policies that were outside the scope of their study but also implemented during the study period.

For the second point, I am not sure what the authors were referring for “functional form of infectious disease."

In order to evaluate the impact of a policy on an infectious disease outcome, we need to make assumptions about how the disease outcome would have changed in the absence of the policy (a counterfactual scenario). We typically encode our assumptions about how this trend would look (e.g. linear or logarithmic) in the choice we make about the functional form of our model. We have edited the language in the manuscript to clarify this point.

7. Page 18, the authors acknowledge the tool “ was limited to a very narrow - albeit critical - set of items.” For the readers to understand this, the authors should consider explaining in more detail what each of these criteria means and how it is linked to the quantitative measure for drawing “causal inference.” Also, it will be helpful to explain how this set of criteria is similar or different from other policy evaluation frameworks.

> **Functional form: “Is the functional form of the *model used for the trend in counterfactual infectious disease outcomes* (e.g., linear, *non-parametric, exponential, logarithmic, etc.,*) well-justified and appropriate?”**
>
> **…**
>
> ***Graphical presentation refers to how the authors present the data underlying their impact evaluation method. This is a critical criteria for assessing the potential validity of the assumed model. The key questions here are whether any chart shows the outcome over time and the assumed models of the counterfactuals. To meet a high degree of confidence in this category, graphical displays must show the outcome and connect to the counterfactual construction method.***

Functional form refers to the statistical functional form of the trend in counterfactual infectious disease outcomes (i.e. the assumptions used to construct counterfactual outcomes). This may be a linear function, non-parametric, exponential or logarithmic function, infectious disease model projection, or any other functional form. The key criteria here are whether this is discussed and justified in the manuscript, and if so, is it a plausibly appropriate choice given infectious disease outcomes.

Timing of policy impact refers to assumptions about when we would expect to see an impact from the policy vis-a-vis the timing of the policy introduction. This would typically be modelled with leads and lags. The impact of policy can occur before enactment (e.g., in cases where behavior change after policy is announced, but before it takes place in anticipation) or long after the policy is enacted (e.g., in cases where it takes time to ramp up policy implementation or impacts). The key criteria here are whether this is discussed and justified in the manuscript, and if so, whether it is a plausibly appropriate choice given the policy and outcome.

Concurrent changes refers to the presence of uncontrolled other events and changes that may influence outcomes at the same time as the policy would impact outcomes. In order to assess the impact of one policy or set of policies, the impact of all other forces that differentially impact the outcome must either be negligible or controlled for. The key criteria here are whether it is likely that there are substantial other uncontrolled forces (e.g. policies, behavioral changes, etc) which may be differentially impacting outcomes at the same time as the policy of interest.

Reviewer: 4

Dr. Peng JIANG, Sichuan University

Comments to the Author:

Comments for bmjopen-2021-053820.

This is an interesting manuscript with a novel perspective for a systematic review of study design and evidence strength under the COVID-19 background. After going through the manuscript, I have some suggestions/comments to improve its quality.

1. As for the limitation, the authors thought that their review tool was limited to a very narrow set of items and the data quality. The reviewed papers were searched by November 2020. The readers might be curious about the overall time-serial variations in reviewed papers related to COVID-19 policies. Actually, the authors have the chance to add some latest papers and analyse the trends along with the COVID-19 development.

While we would love to do so, our sample size does not yield sufficient data to be able to examine trends over time with any reasonable level of certainty. We would further love to have been able to explore more recent studies, but unfortunately are unable to do so at this time

given the several months of review required for this type of study. Our priority at this time is to make these results available to researchers and decision-makers.

2. The “Methods” part is helpful for readers as this manuscript presents some new processes to evaluate the papers using expert experience. However, the “Methods” part can be placed into the appendix considering the length of this manuscript. This can also help readers focus more on the exciting findings/observations.

We prefer to put the methods section earlier in the manuscript as it is important for readers to be able to understand how a study was done before reading the results. Just as we believe the methods to be the most critical component of the papers we are reviewing, we prefer that readers understand and can critically evaluate our methods in depth before the results.

3. In the “Study Selection” part, eight reviewers screened the title and abstract of each article. Why are eight reviewers arranged for such a work? In addition, why not an odd number of reviewers? Generally, an odd number is more practical as the reviewers may offer conflicting opinions.

Apologies for our unclear wording; each article was screened by up to three reviewers, split randomly among 8 people performing the screening. We have clarified this as below.

> ***Two reviewers were randomly selected to screen the title and abstract of each article for the inclusion criteria. In the case of a dispute, a third randomly selected reviewer decided on acceptance/rejection. Eight reviewers participated in the screening.***

4. In the section of “Review tool for data collection”, “Finally, reviewers were asked a summary question: Overall: “Do you believe that the design is appropriate for identifying the policy impact(s) of interest?” This is quite a subjective question. However, the reviewer did not find any mathematical modelling regarding group decisions in this manuscript.

This was, by design, a subjective assessment of the totality of the question. Numerical indices are broadly inappropriate for strength of evidence assessments, and comprehensive subjective assessment is necessary to evaluate strength of evidence as a whole. This is similar to advice from other tools. For example, even the new algorithms for scoring in the new RoB 2 tool from Cochrane only suggest that these algorithms are only “proposed” judgements, requiring that analysts verify or make changes to these as appropriate (https://training.cochrane.org/handbook/current/chapter-08). It is entirely plausible, for example, for a study to meet all but one criteria but for the one remaining to be sufficiently violated that the entire collective category is compromised. Alternatively, there could be many minor violations of all of the criteria, but that they were collectively not sufficiently problematic to impact overall ratings.

For each of the four key questions, reviewers were given the option to select "No," "Mostly no," "Mostly yes," and "Yes" with justification text requested for all answers other than "Yes." Each question had additional prompts as guidance, and with much more detail provided in the full guidance document. ***Ratings are, by design, subjective assessments of the category according to the guidance. We do not use a numerical scoring, for similar reasons as Cochrane suggests that the algorithms for summary judgements for the RoB2 tool are merely "proposed" assessments, which reviewers should change as they believe appropriate. It is entirely plausible, for example, for a study to meet all but one criteria but for the one remaining to be sufficiently violated that the entire collective category is compromised. Alternatively, there could be many minor violations of all of the criteria, but that they were collectively not sufficiently problematic to impact overall ratings. Further, reviewers were also tasked with considering room for doubt in cases where answers to these questions were unclear.***

5. "The citation counts for accepted articles were obtained through Google Scholar on January 11". However, the reviewed papers were not published same time. Therefore, the calculated statistics are not supportive. One option can be the citation counts of those papers within the same duration, e.g. half a year or one whole year after publication.

We agree that it would be more precise to use such a measure. However, we are only using these statistics to describe the broad impact of these studies. Citations counts are never used as a control variable, no between-study comparisons are made, and no associations with it are run.

6. In the "Abstract" part, "outcomes published on November 26 or earlier and screening". Please add the 'year' information, as it may confuse several years after the outbreak of the CIVOD-19 pandemic.

Good catch, thanks! We have added this.

7. In the "Strengths and Limitations" part, "they may make other contributions to the literature." The authors should make the contributions apparent.

Yes, thanks! We have added the following:

These contributions may include providing data for other analyses and purposes, providing methodological proposals for other studies to improve upon, generating actionable evidence via other, non-primary results, or other uses.

VERSION 2 – REVIEW

REVIEWER	Alessandro Rovetta Mensana srls, Research and Disclosure Division
REVIEW RETURNED	30-Sep-2021

GENERAL COMMENTS	Dear authors, thank you for your professional answers and revisions. The paper has substantially improved compared to the previous version. However, I suggest addressing a couple of key points further. 1) Answer 2.1. “Inter-rater reliability (IRR) was assessed using Krippendorff’s alpha.[23,24] Rather than more typical uses intended as an examination of the “validity” of ratings, the IRR statistic in this case is being used as a heuristic indicator of heterogeneity between reviewers during the independent phase, where heterogeneity is both expected and not necessarily undesirable.” Since, as well specified in the manuscript, this is an unconventional way of using an IRR measure, I suggest briefly motivating why "violations of the metric's assumptions would not substantially impact our finding of high heterogeneity." This is a methodologically relevant aspect and could be a reason for citations in other similar papers. 2) Answer 2.2 [Journal Impact Factor]. Dear authors, I totally agree with your answer, and I would like to point out that my comment did not question the methodology of your study. Indeed, precisely for the reasons that you have explained, I strongly suggest motivating in the manuscript the choice to consider this metric (e.g., "This helps readers understand that the included articles are from high-impact, highly visible literature. Impact factor is never used as a control variable, and no associations with it are run.") Indeed, in addition to answering the methodological question "Why was the impact factor considered?", this resolves any potential doubts on a controversial topic and reinforces what is stated in the Discussion section.

REVIEWER	J. Rod Queensland University of Technology
REVIEW RETURNED	20-Sep-2021

GENERAL COMMENTS	I think most of my comments and other author comments have been sufficiently addressed. Therefore, I consider the manuscript of "the best possible quality" that it could possibly be with the current knowledge base about the topic. I would just like the authors to add something similar to the sentence below. This would communicates their own appraisal of the tool more explicitly. I see as a potential location the end of the discussion. (Please apologize me if this is mentioned explicitly but after reading the discussion multiple times I could not find this point being presented explicitly). "this study is merely a first pass for being plausibly actionable, and further and more detailed review is needed for the few studies that passed these minimal criteria. This tool is intended as a minimal set of criteria, rather than a scoring system, similar to recommendations for other tools such as the Risk of Bias". Therefore, "some of our main... recommendations are for the academic community to better vet important studies such that decision-makers have clearer signals of what is ..actionable.. information".

	I feel the authors answered my main concerns but they were not explicitly communicated in the manuscript, which I am assuming is one of the steps necessary for improving the production of "actionable information" for the audience they are aiming. Namely, the academic community. This would perhaps prevent policymakers for giving to much weight on the present tool and use it as "one of the possible lenses" to which they could asses and triangulate the strength of the literature. Thanks again to the authors and the editor for engaging me in this conversation.

REVIEWER	Zhaoli Dai Macquarie University, Australian Institute of Health Innovation
REVIEW RETURNED	07-Oct-2021

GENERAL COMMENTS	Thank you for your efforts in addressing my comments sufficiently. I have no further comments.

VERSION 2 – AUTHOR RESPONSE

Reviewer: 2

Dr. J. Rod, Queensland University of Technology

Comments to the Author:

I think most of my comments and other author comments have been sufficiently addressed. Therefore, I consider the manuscript of "the best possible quality" that it could possibly be with the current knowledge base about the topic.

I would just like the authors to add something similar to the sentence below. This would communicates their own appraisal of the tool more explicitly. I see as a potential location the end of the discussion. (Please apologize me if this is mentioned explicitly but after reading the discussion multiple times I could not find this point being presented explicitly).

"this study is merely a first pass for being plausibly actionable, and further and more detailed review is needed for the few studies that passed these minimal criteria. This tool is intended as a minimal set of criteria, rather than a scoring system, similar to recommendations for other tools such as the Risk of Bias". Therefore, "some of our main... recommendations are for the academic community to better vet important studies such that decision-makers have clearer signals of what is ..actionable.. Information".

I feel the authors answered my main concerns but they were not explicitly communicated in the manuscript, which I am assuming is one of the steps necessary for improving the production of "actionable information" for the audience they are aiming. Namely, the academic community. This would perhaps prevent policymakers for giving to much weight on the present tool and use it as "one of the possible lenses" to which they could asses and triangulate the strength of the literature.

We agree that this is critical information to convey to the reader and have made some changes to make that clearer). We had relegated this to the introduction and methods sections, but have now included this in the discussion as well. To further emphasize these points (as well as a response to another reviewer), we have added the following to the discussion section:

> **"This study was designed to assess a minimal set of criteria critical to the design of impact evaluation studies of COVID-19 policies. Studies found meeting these criteria would require further and more comprehensive review for assessing overall quality and actionability. Unfortunately, exceedingly few studies we reviewed, taken largely from the high-profile literature, were found to meet these minimal criteria."**

The introduction section contains the following: "Second, rather than being a comprehensive review of every possible aspect of what might be considered "quality," this is a review of targeted critical design features for actionable inference for COVID-19 policy impact evaluation and methods. It is designed to be a set of broad criteria for minimal plausibility of actionable causal inference, where each of the criteria is necessary but not sufficient for strong design. Issues in other domains (data, details of the design, statistics, etc) further reduce overall actionability and quality, and thorough review in those domains is needed for any studies passing our basic minimal criteria."

The methods section contains the following "The criteria were designed to establish minimal plausibility of actionable evidence, rather than certification of high quality." and "A "yes" rating does not necessarily indicate that the study is strongly designed, conducted, or is actionable; it only means that it passes a series of key design checks for policy impact evaluation and should be considered for further evaluation. The papers may contain any number of other issues that were not reviewed (e.g., statistical issues, inappropriate comparisons, generalizability, etc.,). As such, this should only be considered an initial assessment of plausibility that the study is well-designed, rather than confirmation that it is appropriate and applicable."

Thanks again to the authors and the editor for engaging me in this conversation.

Thank you for your great comments!

Reviewer: 1

Dr. Alessandro Rovetta, Mensana srls

Comments to the Author:

Dear authors, thank you for your professional answers and revisions. The paper has substantially improved compared to the previous version. However, I suggest addressing a couple of key points further.

1) Answer 2.1. “Inter-rater reliability (IRR) was assessed using Krippendorff’s alpha.[23,24] Rather than more typical uses intended as an examination of the “validity” of ratings, the IRR statistic in this case is being used as a heuristic indicator of heterogeneity between reviewers during the independent phase, where heterogeneity is both expected and not necessarily undesirable.” Since, as well specified in the manuscript, this is an unconventional way of using an IRR measure, I suggest briefly motivating why "violations of the metric's assumptions would not substantially impact our finding of high heterogeneity." This is a methodologically relevant aspect and could be a reason for citations in other similar papers.

Thank you for your comment. We agree that this is not the typical use case for this statistic, noting that we believe it to be an appropriate one given our atypical study design and methods. We further agree that a detailed study and discussion of the impact of specific violations of this statistic’s assumptions would be of value to the literature, noting that other publications (including those cited) provide this information. However, within the scope of this paper, we do not believe that these impact the discussion and conclusions of our analysis, and as such are largely out of scope of this paper. We are also reassured that in this study any given set of reasonable metrics of heterogeneity would show an identical conclusion given the sheer degree of heterogeneity, as corroborated by the additional statistics we have included.

2) Answer 2.2 [Journal Impact Factor]. Dear authors, I totally agree with your answer, and I would like to point out that my comment did not question the methodology of your study. Indeed, precisely for the reasons that you have explained, I strongly suggest motivating in the manuscript the choice to consider this metric (e.g., "This helps readers understand that the included articles are from high-impact, highly visible literature. Impact factor is never used as a control variable, and no associations with it are run.") Indeed, in addition to answering the methodological question "Why was the impact factor considered?", this resolves any potential doubts on a controversial topic and reinforces what is stated in the Discussion section.

We are glad we agree! We have briefly included a mention of this in a new section in the discussion, which we hope resolves both R1 and R2’s comments simultaneously, below:

> **“Importantly, this study was designed to assess a minimal set of criteria critical to the design of impact evaluation studies of COVID-19 policies. Studies found meeting these criteria would require further and more comprehensive review for assessing overall quality and actionability. Unfortunately, exceedingly few studies we reviewed, taken largely from the high-profile literature, were found to meet this minimal criteria.”**

Reviewer: 3

Dr. Zhaoli Dai, Macquarie University

Comments to the Author:

Thank you for your efforts in addressing my comments sufficiently. I have no further comments.

Thank you very much for your great comments!